Lecture 8: Introduction to Z-Transform & FIR Filters

1. Z-Transform & Its ROC

2. Important properties of Z-Transform

3. System Function & Frequency Response / Output of an LTI system for a sin Input

4. Different Types of Filters Based on Freq. Response

5. FIR Filters (System Function & Frequency Response)

1. Z-Transform & Its ROC

A transform especially defined for DT signals.

The Z-transform of $x[n]$ is defined as:

$$\boxed{X(z) = \sum_{n=-\infty}^{+\infty} x[n]\, z^{-n}}$$

we often write the relationship between $x[n]$ & $X(z)$ as:

$$x[n] \xleftrightarrow{Z} X(z) \quad \text{or} \quad X(z) = Z\{x[n]\}$$

We call them a transform pair

We may have coverage issues $(|X(z)| \to \infty)$ and we need to know for which z, the transform $X(z)$ exists $\to$ ROC: The Region of Convergence of $X(z)$ is all z values for which it attains a finite value:

$$\boxed{\text{ROC: All } z \in \mathbb{C} \text{ for which } |X(z)| < \infty}$$

We always need to determine ROC for a Z-transform.

Example 1: $x[n] = \begin{cases} 1 & n=N \\ 0 & \text{otherwise} \end{cases}$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = z^{-N}$$

for z^{-N} → $\begin{cases} \text{if } N>0 \rightarrow z^{-N} = \frac{1}{z^N} \rightarrow \lim_{z\to 0} |z^{-N}| = \infty \\ \text{ROC: All } z \in \mathbb{C} \text{ except } z=0 \\ \text{if } N=0 \rightarrow z^{-N} = z^0 = 1 \rightarrow \text{ROC: All } z \in \mathbb{C} \\ \text{if } N<0 \rightarrow z^{-N} = z^{pr} \rightarrow \lim_{z\to\pm\infty} |z^{-N}| = \infty \\ \text{ROC: All } z \in \mathbb{C} \text{ except for } z = \pm\infty \end{cases}$

Example 2: $x_1[n] = [\underline{1}\ \ 2\ \ 3\ \ 4]$ (we can also represent it using a sequence:

$x[n] = \{\underline{1}, 2, 3, 4\}$)

$$X_1(z) = \sum_{n=\infty}^{+\infty} x[n] z^{-n} = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}$$

→ ROC: $\forall z \in \mathbb{C}$ except for $z=0$

Discussion Time 1: $x_2[n] = [1 \;\; 2 \;\; \underline{3} \;\; 4]$

$$X_2(z) = z^{+2} + 2z + 3 + 4z^{-1}$$

ROC: $\forall z \in \mathbb{C}$ except for $z=0$ & $z=\pm\infty$

2. Important Properties of z-transform

P1: Linearity:

Consider: $x_1[n] \longleftrightarrow X_1(z)$ Then $\rightarrow$

$x_2[n] \longleftrightarrow X_2(z)$

$$a_1x_1[n] + a_2x_2[n] \longleftrightarrow a_1X_1[z] + a_2X_2[z]$$

ROC: At Least Intersection of ROC1 & ROC2

(They can cancel each others effect:

example $\rightarrow$ $x_1[n] = -x_2[n]$)

$a_1 = a_2 = 1$

Because if one goes to infinity then sum $\to \infty$

Proof:

$$x_t[n] = a_1 x_1[n] + a_2 x_2[n]$$

$$X_t(z) = \sum_{n=-\infty}^{+\infty} x_t[n] z^{-n} =$$

$$a_1 \underbrace{\sum x_1[n] z^{-n}}_{X_1(z)} + a_2 \underbrace{\sum x_2[n] z^{-n}}_{X_2(z)}$$

P2: $x[n] \leftrightarrow X(z)$ then $\rightarrow$

$$x[n-k] \longleftrightarrow z^{-k} X(z)$$

ROC = ROC of $X(z)$ except for:

$$\begin{cases} z = 0 & \text{if } k > 0 \\ z = +\infty & \text{if } k < 0 \end{cases}$$

Proof:

$$\tilde{x}[n] = x[n-k] \longrightarrow \tilde{X}(z) = \sum_{n=-\infty}^{+\infty} x[\underbrace{n-k}_{m}] z^{-n}$$

$$= \sum x[m] z^{-(m+k)} = z^{-k} X(z)$$

P3 : $x_1[n] * x_2[n] \leftrightarrow X_1(z) . X_2(z)$

ROC: At least the intersection of ROC1 & ROC2. (example: $\frac{(\cancel{z-1})}{(z-2)} \cdot \frac{1}{\cancel{(z-1)}}$

→ ROC: larger than intersection

P4 $x[n] \leftrightarrow X(z)$ Then $x[-n] \leftrightarrow X(\frac{1}{z})$

ROC: $r_2 < |z| < r_1$ → ROC: $\frac{1}{r_1} < |z| < \frac{1}{r_2}$
of $X(z)$

Proof : $\tilde{x}[n] = x[-n] \rightarrow \tilde{X}(z) = \sum_{n=-\infty}^{+\infty} x[\underbrace{-n}_{m}]$

$$.z^{-n} = \sum_{m=-\infty}^{+\infty} x[m] z^{m} = \sum_{m=-\infty}^{+\infty} x[m] (z^{-1})^{-m}$$

$$= X(z^{-1})$$

3. System Function & Frequency Response

$$x[n] \rightarrow \boxed{h[n]} \rightarrow y[n] \qquad \boxed{H(z) = \mathcal{Z}\{h[n]\}}$$

is called **System Function**

For an LTI system:

$$y[n] = x[n] * h[n]$$

$$\xleftrightarrow[\& \, P_3]{z\text{-transform}} Y(z) = X(z) \cdot H(z)$$

$$\rightarrow \boxed{H(z) = \frac{Y(z)}{X(z)}}$$

For an LTI system, Difference Equation is

Example: $y[n] = y[n-1] + 2x[n] - 2x[n-1]$

$$\xrightarrow[\& P_1]{z\text{-transform}} Y(z) = \mathcal{Z}\{y[n-1]\} + 2X(z) - 2\mathcal{Z}\{x[n-1]\}$$

$$\xrightarrow{P_2} Y(z) = z^{-1}Y(z) + 2X(z) - 2z^{-1}X(z)$$

$$\rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{2(1-z^{-1})}{1-z^{-1}} = 2$$

If $|z| = 1$ is in ROC of $H(z)$; then

Frequency Response is defined as:

$$\boxed{H(e^{j\omega}) = H(z\big|_{z=e^{j\omega}})}$$

Point : For an LTI system, if

$$x[n] = A\cos(\hat{\omega}_0 n + \phi)$$

Find the output based of the FR of the System?

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} h[k]\,x[n-k]$$

$$; \; x[n] = A\left(\frac{e^{j\hat{\omega}_0 n + j\phi} + e^{-j\hat{\omega}_0 n - j\phi}}{2}\right)$$

$$y[n] = \sum h[k] \cdot \frac{A}{2}\left(e^{j\hat{\omega}_0(k-n) + j\phi} + e^{-j\hat{\omega}_0(k-n) + j\phi}\right)$$

$$\rightarrow y[n] = \frac{A}{2}\underbrace{H(e^{-j\hat{\omega}_0})}_{|H(e^{+j\hat{\omega}_0})|\, e^{-j\angle H(e^{j\omega})}} e^{-j(\hat{\omega}_0 n + \phi)} + \frac{A}{2} H(e^{j\hat{\omega}_0})\, e^{j(\hat{\omega}_0 n + \phi)}$$

$$\boxed{y[n] = A\,|H(e^{j\omega_0})|\cos[\hat{\omega}_0 n + \phi + \measuredangle H(e^{j\omega})]}$$

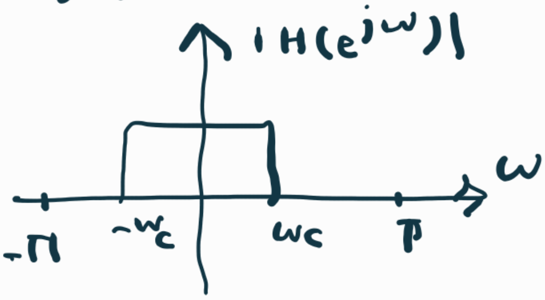
So, the output is also a real sinusoidal signal that has been affected by the magnitude and phase of the frequency response of the system

4. Different Types of Filters Based on the Freq. Response:

Point: as $e^{j\omega + 2\pi} = e^{j\omega} \rightarrow H(e^{j\omega})$ is periodic with the period of 2π

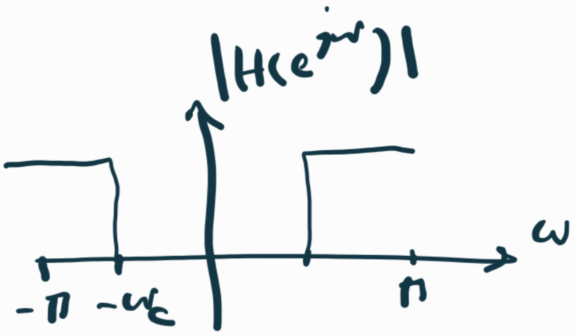
high frequencies $= \pm\pi$ Low freq $\rightarrow 0$

Consider the input of $\sum_k A_k \sin(\omega_k n + \alpha_k)$

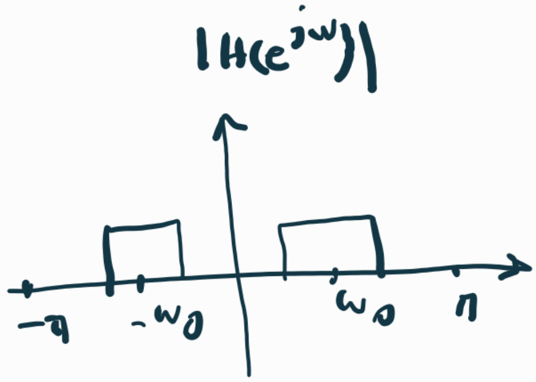


Low-Pass (LP)

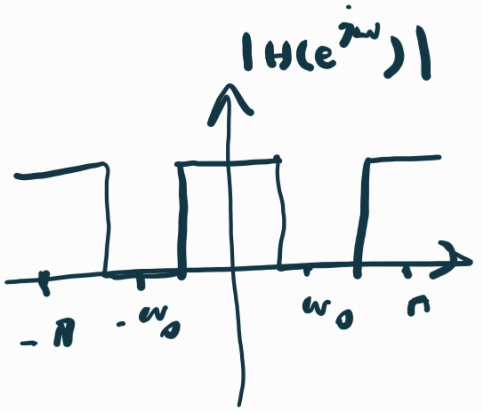
$\rightsquigarrow$ only passes the low frequencies of $x[n]$



High_Pass (HP)



Band-Pass (BP)



Band_stop (BS)

5. FIR Filters

For FIR filters: $y[n] = \sum_{k=0}^{M} b_k x[n-k]$

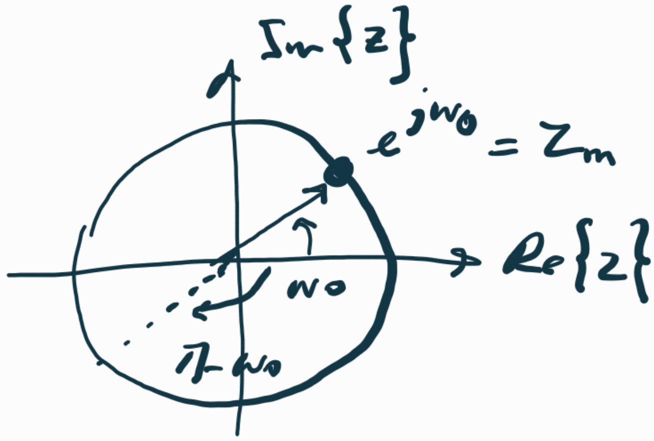
$$\xrightarrow[P_1 \& P_2]{\text{Z\_transform}} Y(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$\longrightarrow H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{M} b_k z^{-k} = \frac{\sum_{k=0}^{M} b_k z^{M-k}}{z^M}$$

$$= G \frac{(z-z_1) \cdots (z-z_M)}{z^M} \rightarrow \begin{cases} M \text{ poles at } z=0 \\ M \text{ zeros at } \{z_1, \dots z_M\} \end{cases}$$

$$\rightarrow |H(e^{jw})| = |H(z=e^{jw})| = G \frac{|e^{jw}-z_1| \cdots |e^{jw}-z_m|}{e^{jwm}}$$

If we have a zero at e^{jw_0} (on unit circle)



at $w = w_0 \rightarrow |H(e^{jw_0})| = 0$

as $|e^{jw_0} - e^{jw_0}| = 0$

at $w = \pi - w_0 \rightarrow |e^{jw} - e^{jw_0}|$ has the max value.

Examples $\longrightarrow$ Slides.